

Samuel Lima Braz

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[Portfolio](#) | [GitHub](#) | [Hugging Face](#)

PROFILE

Computer engineering graduate who trains, evaluates, and serves vision models, then runs them on cameras and boards. Three and a half years on autonomous drones at Black Bee Drones, mostly indoor without GPS, and nearly two years of industry ML on datasets, model comparison, and serving. Looking for computer vision and robotics roles where the model has to work on a moving machine.

SKILLS

Computer vision	Object detection, instance segmentation, classification (YOLO, DETR, RF-DETR); OpenCV (Hough, RANSAC, ArUco)
Depth and SLAM	RGB-D cameras (RealSense), visual SLAM, VIO, EKF fusion (ArduPilot, PX4), depth-based obstacle avoidance
Robotics	ROS 2, ArduPilot, PX4, MAVLink, MAVROS, Gazebo SITL, Isaac ROS, PID control, state machines, Pixhawk
ML and deployment	PyTorch, synthetic data, Optuna, PEFT (LoRA, QLoRA), TensorRT, ONNX, OpenVINO, NVIDIA Triton, Jetson Orin
Programming	Python, C++, C, Java; Linux, Git, Docker, GitHub Actions (CI/CD), Weights & Biases, MLflow
Languages	English (fluent), Portuguese (native)

EXPERIENCE

Robotics Engineer (Volunteer) — Black Bee Drones, UNIFEI

Apr 2023 to Present

- IMAV 2023 indoor 3rd and Special Achievement Award for highly automated MAV operation (only team to complete it fully autonomously). IMAV 2025 indoor 3rd (wide gate tunnel, landing on a moving platform). SAE Eletroquad 2026 2nd. Also competed in CBR.
- Lead developer of **Nectar SDK**, the ROS 2 kit the team has used as its shared software since 2024. One flight interface for ArduPilot and PX4 (MAVROS, MAVLink, uXRCE-DDS), plus Bebop and Crazyflie; camera factory (RealSense, OAK-D, USB); detection, segmentation, and classification. Docker for x86_64 and ARM64. [Docs](#)
- Indoor navigation without GPS: VIO with Intel RealSense T265 on a Raspberry Pi, then vSLAM with Isaac ROS and RealSense D435i on a Jetson Orin Nano, into ArduPilot EKF3 and PX4 EKF2. Wrote a ROS 2 C++ bridge ([vision_to_mavros](#)) from T265 pose to MAVROS, including ENU/NED conversion and covariance.
- Perception and control on the vehicle: OpenCV colour and line filters, then YOLO detection and instance segmentation converted with TensorRT, OpenVINO, and ONNX. Mission state machines; PID in body, world, and takeoff frames; obstacle pause from D435i depth; same missions in Gazebo SITL before flight. Pixhawk with cameras, GPS, LiDAR, motors, and ESCs.

Machine Learning Engineer Intern — Chunkr (US, remote)

Jun 2025 to Jun 2026

- Built a synthetic document-layout generator from statistics of real pages (positions, co-occurrence, size). Element pool in SQLite with vectorized search and multiprocessing. [Article](#)
- Pretrained YOLO and RF-DETR on 500K synthetic pages mixed with real data, then fine-tuned. After training: per-class confidence filters, box refinement from pixel density, and a merge that raised F1 on every class. A customer layout model went from 0.52 to 0.71 test F1.
- Released chunkr-layout-1 at 80.9% mAP@50 on 1,013 annotated documents. Fine-tuned LayoutLMv3-Large for reading order: 88.1% exact match and 98.2% Kendall's τ on the open [benchmark](#) (733 documents). Served models on NVIDIA Triton; workers on Redis.

Machine Learning Engineer — Tech4Humans

Feb 2025 to Jul 2025

- Open-source handwritten-signature detector on Hugging Face (50M+ downloads). Compared YOLO (v8–v12), DETR, and YOLOs; kept YOLOv8s. Public code, weights, data, and report ([blog](#), [collection](#)).
- Optuna search: +7.94 percentage points F1; recall 76.2% to 91.2%. Triton with ONNX/OpenVINO, 172 ms on CPU.
- Fine-tuned open VLMs ($\leq 8B$) for fields on Brazilian documents with LoRA and QLoRA (Unsloth, MS-Swift), evaluated with vLLM, and served JSON extraction on GPU. Experiment tracking in Weights & Biases and MLflow.

AI Engineer Intern — Tech4Humans

Aug 2024 to Jan 2025

- Compared PEFT (LoRA, QLoRA, IA3) with full fine-tuning on dialogue summarization: QLoRA trained less than 1% of parameters, used 55% less memory, and scored best on every automatic metric. Published on the [Hugging Face blog](#) and [AI News Brazil](#).

PROJECTS

Quantum Assistant — Thesis: synthetic multimodal data and PEFT on Qwen3-VL-8B for Qiskit (circuit diagrams, Bloch spheres, histograms). 8,366 samples; +11–17 pp on Qiskit HumanEval. Published in *Expert Systems with Applications* (Elsevier). [GitHub](#) | [Article](#)

CafeDL — Deep learning library from scratch in Java: hand-written forward and backward passes for Conv2D, max pooling, dense, and dropout layers, 8 optimizers (SGD to Adam), losses, and a training loop with early stopping. No autograd; ND4J only for array math. Trained a QuickDraw sketch classifier used in a JavaFX game. [GitHub](#)

EDUCATION

Bachelor's degree in Computer Engineering, UNIFEI

Mar 2022 to Dec 2025

Technical High School in Industrial Automation, IFMG

Jan 2019 to Feb 2022